

Learning

The knowledge graph from counting to coin sums, the breakdown of every knowledge component in it, and the one lesson built from it — how it teaches, and how it detects what a student got wrong.

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The knowledge graph

From “knows nothing” to “can total a handful of coins.”

The graph is a chart with a forked focus on two concepts that come together to achieve the goal. It begins with concrete steps related to building out a child’s numeral and counting knowledge. On one side of the fork the child builds out arithmetic skill that leads to adding four two-digit numbers — a Common Core standard for second graders. On the other side of the fork is building out new vocabulary, mathematical skills, and applying counting strategies to currency, ending with the ability to price a set of coins of one denomination and to sort a mixed handful into such sets. Those two halves together achieve the final goal: summing a handful of coins.

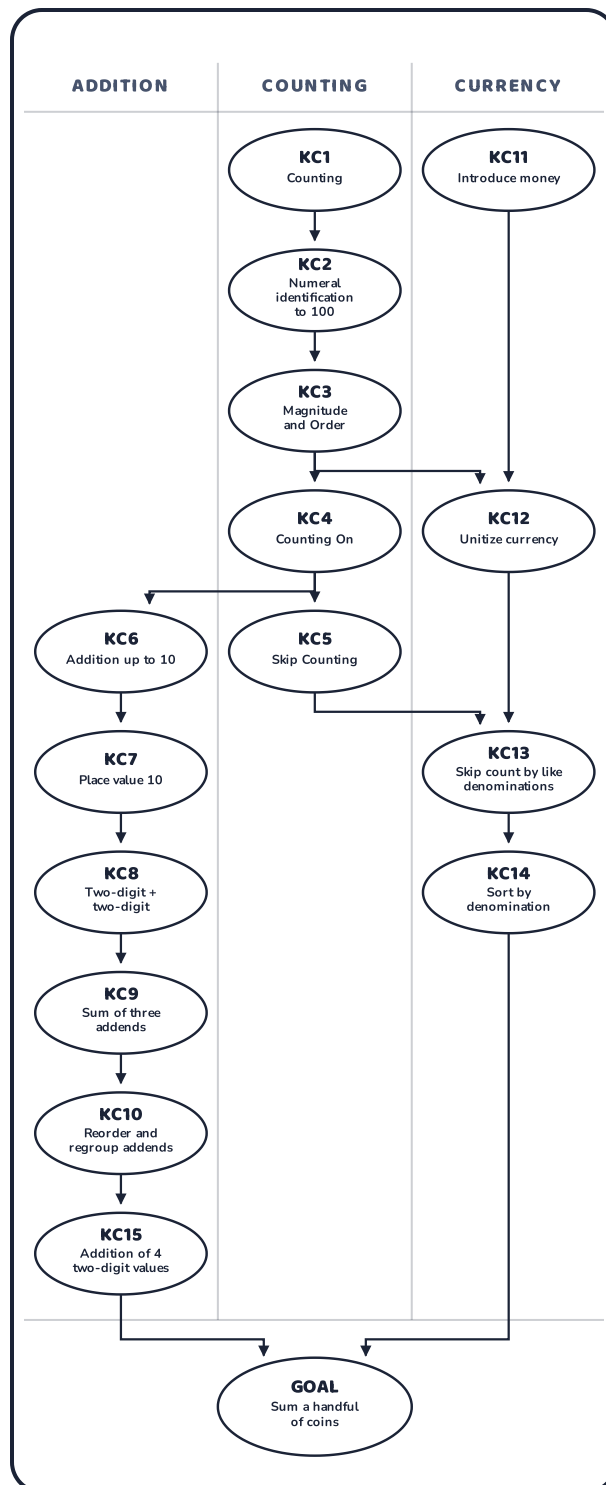
Reading it

Three lanes run top to bottom, and an edge means “you need this first.” **Counting** is the spine: KC1 through KC5. **Addition** forks off KC4 (counting on) and runs to KC15. **Currency** has its own root — KC11, introducing money and its vocabulary, depends on nothing, because naming a penny needs no arithmetic. Two edges cross from Counting into Currency and they are the load-bearing ones: KC3 (magnitude and order) feeds KC12, so that a nickel being physically bigger than a dime while being worth less has somewhere to land; and KC5 (skip counting in the abstract) feeds KC13, where skip counting is put to work on money. KC14 and KC15 converge on the goal.

The path not taken. Another route through this graph exists: drop the addition lane and instead build the ability to skip count across *unlike* denominations — count the quarters, then continue by dimes, then nickels, then pennies, to arrive at the answer in one running count. That strategy is valid and it is faster to teach, but it is limited in the long term: it produces a child who can price coins and no child who can add. It survives in the graph as a supplemental skill, off the critical path. The path shown here uses the deeper mathematical foundation to reach the same result.

On grain size. The brief asks for roughly ten to fifteen nodes, so a few nodes are compacted. KC2 is the clearest case: “numeral identification to 100” is several lessons and several milestones in delivery — numerals to 10, then to 20, then to 100 — not one. The compaction is a presentation choice, not a claim that it is one teachable step.

Dependencies, top to bottom



Fifteen knowledge components in three lanes. An arrow points from a prerequisite to what it unlocks. KC13, shaded in the table overleaf, is the component the lesson teaches.

The knowledge components

One small, teachable piece each, with the standards and research each is answerable to.

Codes resolve in the sources at the end. **CCSS** codes are Common Core; **TEKS** is Texas, cited wherever it splits coin work into steps Common Core leaves implicit; **VA SOL** is Virginia. **LT** lines are developmental levels from the Clements & Sarama learning-trajectories work, each mapped by their team to the standards beside it.

KC	CONCEPT	DESCRIPTION	STANDARDS & RESEARCH
KC1	Counting COUNTING	the student must be able to count discrete objects starting at 1	K.CC.A.1 K.CC.B.4a K.CC.B.5 LT Corresponder → Counter (Small Numbers)
KC2	Numeral identification to 100 COUNTING	the student must be able to assign a numeral to a quantity (e.g. $\square\square = 2$ and $\square\square\square\square = 4$) and to read, write, and sequence numerals through 100. Compacted for brevity: in delivery this is several lessons and milestones — numerals to 10, then to 20, then to 100 — not one.	K.CC.A.3 K.CC.A.1 1.NBT.A.1 LT Counter and Producer (10+) → Counter to 100
KC3	Magnitude and Order COUNTING	the student must be able to define which numerals are bigger/smaller ($5 < 10$, important for nickel dime clarification)	K.CC.C.6 K.CC.C.7 1.NBT.B.3 LT Counting Comparer (10) → Place Value Comparer
KC4	Counting On COUNTING	the student must be able to start from one value and continue counting on from there by 1s	K.CC.A.2 1.OA.C.5 LT Counter from N → Counter on Keeping Track
KC5	Skip Counting COUNTING	the student must be able to count by fixed increments (5s and 10s)	2.NBT.A.2 1.NBT.B.2c LT Skip Counter by 10s to 100 → Skip Counter
KC6	Addition up to 10 ADDITION	the student must be able to sum numbers to 10 fluently — from known combinations or by counting on from one addend, not by counting all from 1	K.OA.A.2 K.OA.A.5 1.OA.C.6 LT Counting Strategies +/- (the level above Find Result, which is counting all)
KC7	Place value 10 ADDITION	the student must develop a schema around the 10s place.	1.NBT.B.2a 1.NBT.B.2b 1.NBT.B.2c K.NBT.A.1 LT Composer with Tens and Ones
KC8	Two-digit + two-digit ADDITION	the student must understand the concept of carrying from the ones to the 10s place when adding two two-digit numbers ($35 + 27$)	1.NBT.C.4 2.NBT.B.5 2.NBT.B.9 LT Deriver +/- → Multidigit +/-

KC	CONCEPT	DESCRIPTION	STANDARDS & RESEARCH
KC9	Sum of three addends ADDITION	the student must be able to add three whole numbers in a single problem, holding a running total across more than one join	1.OA.A.2 <i>The trajectory alignment attaches no level to 1.OA.A.2 directly; it sits between Counting Strategies and Part-Whole.</i>
KC10	Reorder and regroup addends ADDITION	the student must understand that addends may be added in any order and grouped in any way, so a sum can be rearranged into easier pairs (25 + 30 + 15 + 4 worked as 25 + 15 + 30 + 4)	1.OA.B.3 <i>1.OA.B.3 is the permission to reorder; 2.NBT.B.6 then asks for “properties of operations” — this node is that half of it.</i> LT Numbers-in-Numbers +/-
KC11	Introduce money CURRENCY	the student must be able to identify real coin objects by name (penny, nickel, dime, quarter) and use the terminology of cents. This introduces money and its vocabulary before any worth is attached to a particular coin.	TEKS K.4(A) VA 1.NS.1(F) <i>TEKS K.4(A) is “identify U.S. coins by name” — a full grade before value. CCSS has no money standard before grade 2.</i>
KC12	Unitize currency CURRENCY	the student must be able to identify the worth of a coin in cents: 1 cent = penny, 5 cent = nickel, 10 cents = dime, 25 cents = quarter	2.MD.C.8 TEKS 1.4(A) <i>TEKS 1.4(A) is the same verb as K.4(A) but “by value, and describe the relationships among them.”</i>
KC13	Skip count by like denominations CURRENCY	the student must be able to skip count to arrive at the value of a set of coins of the same denomination.	2.NBT.A.2 2.MD.C.8 TEKS 1.4(C) <i>TEKS 1.4(C) states this node almost verbatim: “count by twos, fives, and tens to determine the value of a collection of pennies, nickels, and/or dimes.”</i>
KC14	Sort by denomination CURRENCY	the student must be able to sort a mixed handful of coins into groups of like denomination themselves, producing the like-denomination sets that KC13 knows how to price	<i>Strategy node — no standard states it. It is the step that turns one hard mixed-unit count into four easy single-unit counts, and the precondition TEKS 2.5(A) (“value of a collection of coins up to one dollar”) silently assumes.</i>
KC15	Addition of 4 two-digit values ADDITION	the student must be able to derive the sum of four two digit numbers (up to 100). This is required for summing up the values quarters, dimes, nickels, and pennies.	2.NBT.B.6 <i>Verbatim: “Add up to four two-digit numbers using strategies based on place value and properties of operations.”</i> LT Problem Solver +/-

The lesson

KC13 — skip count by like denominations.

The lesson takes one node and teaches it completely: given a pile of coins that are all the same denomination, find what the pile is worth. It is the node where the two halves of the graph first touch money, and it is the node that makes the rest of the currency lane possible — KC14 (sorting a mixed handful) only pays off if you can price the piles it produces.

Assumed mastered

Its two direct prerequisites, and their ancestors:

- **KC12** the worth of each coin in cents
- **KC5** skip counting by 5s and 10s in the abstract
- and behind those, **KC11** coin names and the word “cents,” plus **KC1** **KC3** counting, numerals, and order

Nothing else. The lesson never asks the student to add.

Deliberately out of scope

- **KC14** sorting a mixed handful — every pile in this lesson is already one denomination
- **KC15** adding the four subtotals
- the goal node itself

Those are the next lessons. This one buys the strategy they depend on.

Why skip counting, and why it is taught this way

Skip counting by the denomination of a coin does two jobs at once. It re-enforces the value attributed to each denomination — you cannot count a pile of nickels by 5s without rehearsing that a nickel is 5 cents, thirty times over — and it introduces the strategy the capstone runs on: **count by denomination first, then sum the totals**. A child who reaches KC15 having never met that strategy is a child staring at a mixed handful with no handle on it.

The instruction is built around separating two quantities that a struggling student tends to fuse: **what one coin is worth** and **how many coins there are**. They are introduced one at a time, in that order, and only then brought together on the skip-count grid — a hundred chart the count physically walks across. The grid is what makes the product visible without requiring multiplication: counting by 5 four times *lands somewhere*, and the place it lands is the answer.

Three modes, in order

Each hands over only when the student has an answer marked right. The shape is the same in all three; what changes is who does the work.

- 1 Introduce the task.** The lesson names what is about to happen, so the student is not inferring the game from the first move.
“We are going to skip count by coins.”
- 2 Focus on the worth of the coin.** The pile is named, the value is recalled, and the step size for the count is stated as a consequence of it.
“These are nickels. Remember nickels are worth 5 cents. We are going to count by 5s.”
- 3 Focus on the number of coins.** Only now does quantity enter, and it is framed explicitly as the number of *times*, not the answer.
“Let’s count the number of nickels, so we know how many times to count by 5.”

4

Apply both to the grid. The two facts are combined into the count, and the cell it ends on is read off as the worth of the whole pile.

“Now let’s count by 5, 3 times.”

Then the count itself, one coin at a time, with the running total called out as each one lands — “5... 10... 15” — and only after it the sentence that names what just happened.

“Three nickels are worth 15 cents.”

Instruction plays those beats on a pile of three nickels: it narrates, boxes the pile, turns the coins over, counts them, walks the chart calling out the running total a coin at a time, and fills the answer in. The student presses Check Answer. **Guided** runs the same shape on four dimes with the doing handed over — each beat waits for the taps it asked for before the next line is spoken: tap a coin to turn it over and check its worth, tap each coin to count the pile, then skip count the chart by eye and type where you landed. **Free play** is ten piles, one at a time, no scaffolding.

Four details that carry weight

The count is said, not described. The chart walk in instruction has no narration line of its own. Each coin lands and the cell it lands on is spoken — “5,” “10,” “15” — which is the count happening out loud rather than a sentence about the count. The coin and the number are one statement, and the pause between them is a floor rather than a cadence, so the student can say the numbers along with it and a slow voice pushes the next coin out instead of being cut off by it.

A coin never carries two numbers at once. The ordinals written under the pile while it is being counted (1, 2, 3) are wiped before the chart is walked, and again before the student types in guided mode. A coin labelled both “3rd” and “worth 15” is the exact confusion skip counting exists to clear up, and a pile still labelled 1 to 4 in front of an empty answer box is an invitation to type 4. The first error type overleaf is the one this guards against, designed out of the screen rather than only caught after the fact.

The answer is staked before it is judged. A typed number goes onto the chart first as an outline star — the student’s claim — and only then does the count run to meet it or miss it. Marking the claim on the same surface the count walks is what makes a wrong answer legible to the student instead of merely red: they can see how far off they were and in which direction, which a red box cannot say.

Narration is generated from the problem, never typed out. Every line is a function of the pile in front of the student, so changing which pile the lesson opens on cannot leave the narration saying “nickels” over a heap of dimes. Values are spelled “5 cents,” not “5¢,” because this text is read aloud at least as often as it is read.

The problem sequence

Two taught, ten tested. Every value is derived from the pile, never hand-typed.

#	MODE	PILE	COUNTED BY	WORTH	SUB-SKILL
p1	Instruction	3 nickels	5s, 3 times	15¢	nickels by 5
p2	Guided	4 dimes	10s, 4 times	40¢	dimes by 10
p3	Free play 1	5 nickels	5s, 5 times	25¢	nickels by 5 — BRIDGE A
p4	Free play 2	1 quarter	25s, once	25¢	quarters by 25 — BRIDGE A
p5	Free play 3	5 dimes	10s, 5 times	50¢	dimes by 10 — BRIDGE B
p6	Free play 4	2 quarters	25s, twice	50¢	quarters by 25 — BRIDGE B
p7	Free play 5	7 pennies	1s, 7 times	7¢	pennies by 1
p8	Free play 6	10 dimes	10s, 10 times	100¢	dimes by 10
p9	Free play 7	15 nickels	5s, 15 times	75¢	nickels by 5 — BRIDGE C
p10	Free play 8	3 quarters	25s, 3 times	75¢	quarters by 25 — BRIDGE C
p11	Free play 9	8 dimes	10s, 8 times	80¢	dimes by 10
p12	Free play 10	7 nickels	5s, 7 times	35¢	nickels by 5

The quarter bridge

Twenty-five is the hard step. A child who can count by 5s and by 10s has those rhythms from the counting lane already; nobody arrives fluent in counting by 25s. So every quarter problem is preceded by a problem with the same answer reached the easy way:

- **Bridge A** — five nickels are 25 cents (p3), and then one quarter is 25 cents (p4). The equivalence is handed to the student the moment it is useful.
- **Bridge B** — five dimes are 50 cents (p5), then two quarters are 50 cents (p6). The same move one place up.
- **Bridge C** — fifteen nickels are 75 cents (p9), then three quarters are 75 cents (p10). The longest count in the lesson, cashed in immediately for the shortest. A student who has taken the strategy on can see that the second pile is the first pile, and the fifteen-coin count they just ground out is what buys them the three-coin one.

All three run the same way round — the easy count first, the quarters second — so the equivalence always arrives as a reward for work already done rather than as a rule to be taken on trust.

What the ten problems cover

Three nickel problems, three dime, three quarter, one penny. Pennies get one because counting by 1s is the degenerate case of skip counting — it is in the set so the student meets all four coins, not because it exercises the skill. Pile sizes run from 1 to 15 so that a student who can only count small piles is separated from one who has the strategy: fifteen nickels cannot be got by counting the coins and hoping. The chart stops at 100 and p8 (ten dimes) lands exactly on its last cell; an answer above 100 is caught before the count runs, and the student is told the chart does not go that far rather than being marked wrong.

Remediation — detecting what went wrong

The brief asks for detection, not the cure. Every wrong answer is classified into an error type and attributed to a sub-skill, and the lesson's output is a report of which sub-skills are shaky.

Three mistakes leave a clean arithmetic signature in the typed answer, and those three are separable without asking the student a single follow-up question. Everything else lands in a fourth bucket that a bare number cannot split any further — which is a limit of the evidence, not of the taxonomy, and the section after the table says what would lift it.

MISTAKE	WHAT IT LOOKS LIKE	HOW IT IS DETECTED	WHERE IT POINTS
Counted the coins, not their worth COUNTED-COINS-NOT-VALUE	Four nickels answered as 4 . The student counted objects and never attached a value to them.	The answer equals the number of coins. <code>answer === count</code>	KC12 unitize currency. The counting is fine; the coin is not yet a unit worth something.
Misattributed the coin's worth WRONG-DENOMINATION-VALUE	Four nickels answered as 40 — a dime's step applied to a nickel. The method is right; the constant is wrong.	The answer divides evenly by the number of coins, and the step that implies is some other real coin's value. <code>answer % count === 0 && implied !== step && coinByValue(implied)</code>	KC12 , and where the confusion is nickel-for-dime, KC3 — the bigger coin being worth less.
Miscounted the coins MISCOUNTED-COINS	Four nickels answered as 15, 25 or 45 — counted by 5s three times, five times, or nine.	The answer is a whole number of this coin's steps, but not the right number of them. <code>answer % step === 0</code>	KC1 counting — keeping track of how many, and holding on to the last number said. The coin knowledge is intact.
Lost the count, or lost the number on the way to the box TRANSPOSED-DIGITS / UNPARSEABLE / UNKNOWN	Four nickels answered as 19 — one uneven step somewhere in the count. Or the chart landed on 45 and the student typed 54 , or typed nothing usable: the mathematics was done and then dropped between the grid and the keyboard.	Whatever the first three signatures do not claim. Digit transposition and an unusable entry are named where they fit; the rest is recorded with the denomination, so the real distribution can be read off later rather than guessed at now.	Not yet decided. A broken count points at KC5 ; a slip between grid and keyboard points nowhere in the graph, and reporting it as a maths error would be a lie about the student. From the typed number alone the two are indistinguishable.

Order of checks

The classifier tries the signatures in a fixed order — exact match, unusable entry, count-as-answer, wrong step, miscount, transposition — because some answers satisfy two tests at once and the tie has to be broken deliberately. Two nickels answered as 20 is both a whole number of 5s, four of them, and a dime's step taken twice. The second is the more specific claim and the more likely child: the method is intact and the coin's worth is not, which is a different lesson from losing count. So the wrong step is checked first — but it only claims an answer when the step it implies is a real coin's value. Five nickels answered as 20 implies a 4-cent step, and there is no 4-cent coin, so that one falls through and is read as a miscount.

Splitting the fourth bucket — a proposed supplemental step

Four nickels answered as **19** is the case that defeats a typed number. It could be a count that stepped by 4 once somewhere in the middle — a real failure of KC5, and a teachable one, because the child is skip counting but cannot hold the step steady. Or it could be a chart walked correctly to 20 and mistyped on the way to the box. Same number, two different children, and nothing in the answer separates them.

The missing evidence is the count itself, and the chart is already the place it happens. A supplemental step would **require the student to mark their count on the skip-count grid before the answer box opens** — tap the cells they land on, then type where they finished. That yields two records where there was one, and the fourth bucket splits three ways:

- **The marked cells are uneven or wrong.** The skip count broke, and the grid says *where* it broke and by how much — which coin, which step. That is a **KC5** finding with a location attached, far more useful than “wrong.”
- **The cells are right, the typed number is not the last one.** A transfer failure, identified rather than inferred, and correctly reported as no maths error at all.
- **The cells are right, the number matches, and both are wrong.** They counted a correct sequence the wrong number of times — which corroborates the miscount signature instead of leaving it as an arithmetic coincidence.

It is worth naming what this costs. Guided mode deliberately does *not* have the student tap the chart, because landing the coins for them is doing the counting on their behalf. This step is the opposite move — it records a count the student has already made rather than walking them through one — but it still puts a step between the thinking and the answer, and a child who had it right the first time gains nothing from it. So it belongs as a *second* attempt rather than a first: on a wrong answer, the problem returns with the grid live and the instruction to show the count. The student who was right never meets it, and the student who was wrong produces exactly the evidence needed to say why.

One blind spot that stays, stated rather than papered over. On p7 the number of pennies and the value coincide — seven pennies are 7 cents — so a student who counts objects instead of value produces the right answer and is invisible to the first check. That is true of any penny pile, whatever its size: the step is 1, so counting the coins and pricing them are the same arithmetic. Pennies cannot diagnose that mistake; only the other three denominations can, which is a further reason the penny problem is one of ten and not three.

The report

The ten free-play problems are the mastery check. A student who gets one wrong is not stopped — they answer again until the problem is right, because leaving a six-year-old stuck on a screen teaches nothing — but it is the **first** attempt on each problem that is classified and scored. At the end, the attempts roll up into a report: how many were right, a count of each error type, and a count of misses per sub-skill sorted worst first.

That last list is the deliverable. It says *where* the student needs help in the language of the graph — “three misses on quarters by 25, none anywhere else” points at the bridge, not at the child — and it is the hand-off point for remediation, which this brief puts out of scope.

Sources

Codes in the tables above resolve here.

- **CCSS** — Common Core State Standards for Mathematics, grades K–2: K.CC, K.OA, 1.OA, 1.NBT, 2.OA, 2.NBT, 2.MD. Money appears exactly once in K–2, at 2.MD.C.8, which is why the currency lane leans on other states for its intermediate steps.
- **TEKS** — Texas Essential Knowledge and Skills, mathematics: §111.2 Kindergarten, Grade 1, Grade 2. Cited because TEKS splits coin work into steps Common Core leaves implicit — identify by name in kindergarten, identify by value in grade 1, count a collection by twos, fives and tens in grade 1.
- **VA SOL** — Virginia 2023 Mathematics Standards of Learning; 1.NS.1(f) names coin identification “by their attributes.”
- **LT** — Clements & Sarama, *Learning and Teaching with Learning Trajectories* (Marsico Institute, University of Denver), with their alignment of every developmental level to CCSS.

Playcademy · Coin Sums · LEARNING.pdf — the knowledge graph, the KC breakdowns, and the lesson flow plus remediation. Companion documents: FRONTEND.pdf and MULTIPLAYER.pdf.